Key features of parabolas

1

a Downwards **b** Upwards

2 Upward

3 No, because the entire parabola will be above the x-axis. Also, there is no solution for the equation $x^2 + 6 = 0$.

4

a No **b** Yes **c** No **d** Yes

5

a $x = 0, 4$ **b** $y = 0$ **c** 4

6

a Concave up **b** $y = 9$ **c** $y = 1$ **d** $x = -2$

e $x < -2$

7

a $(-2, 0)$ **b** Vertex: $(-2, 0)$

c

i False **ii** True **iii** True **iv** True

8

a 2 **b** 1

9 Concave up

10

a $(-4, 2)$ **b** 2

11 $x = 3$

12

a No **b** No **c** Yes

13 Quadratic function A

14 $x = 0$

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15 $y = 25$

16

a $x = 14, 6$ **b** $x = 10$ **c** Concave down **d** $y = 16$

17 The x-coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y-coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x.

18 $x = \frac{1}{4}$

19 $x = -3, 2$

20

a False **b** True

21 $x = \frac{-3 \pm \sqrt{93}}{6}$

Graphing quadratics

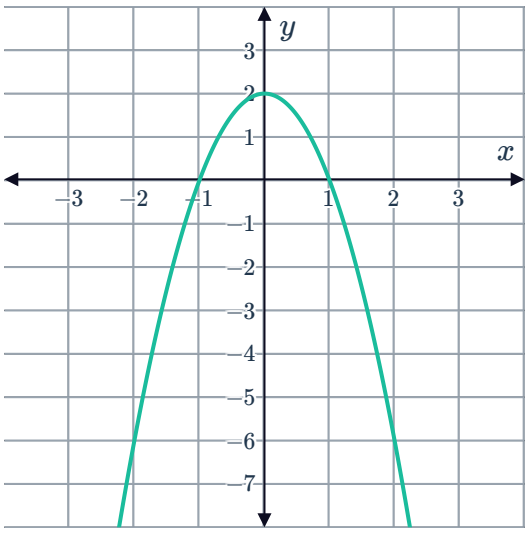
22

a Concave down

b More steep

c $(0, 2)$

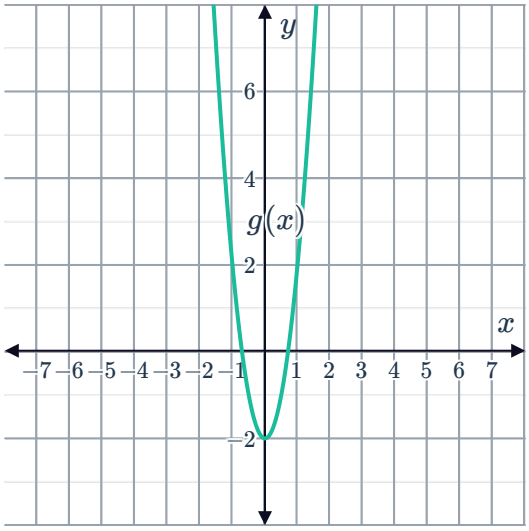
d



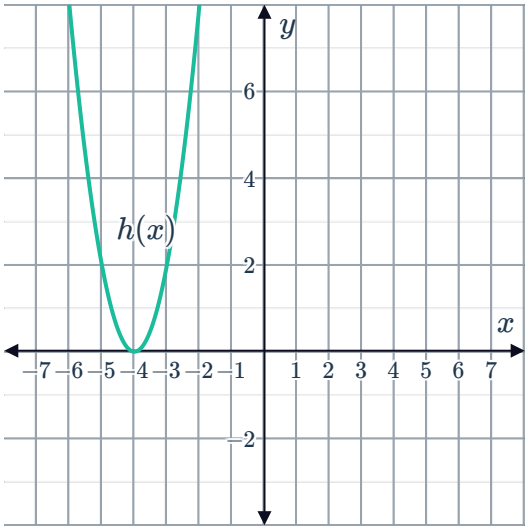
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23

a



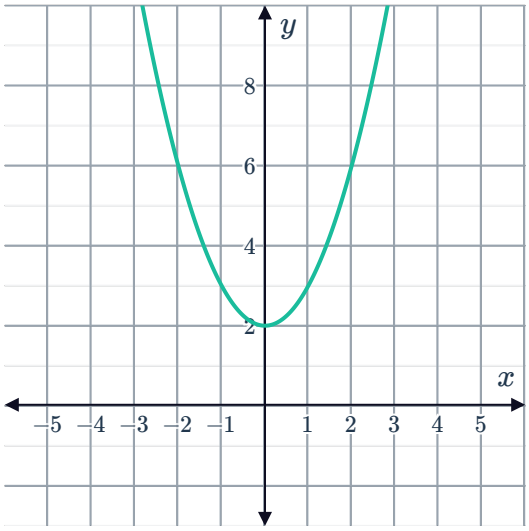
b



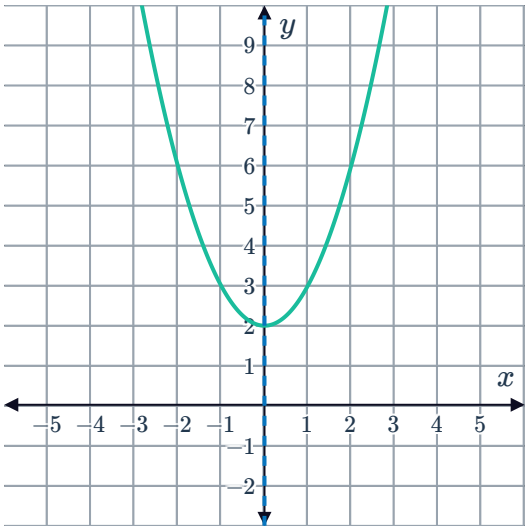
24 $y = -x^2 + 5$

25

a



b

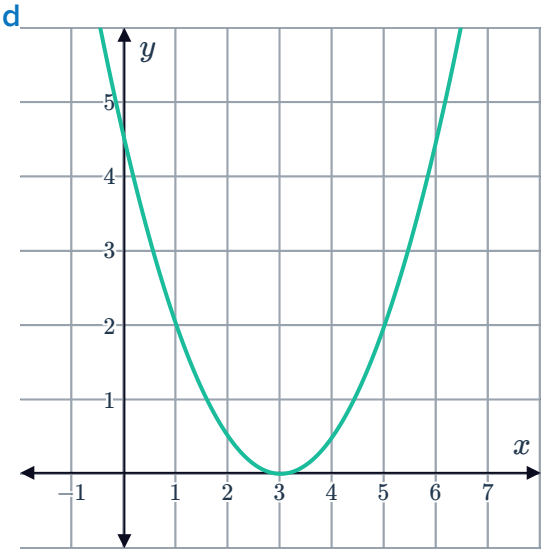


c $(0, 2)$

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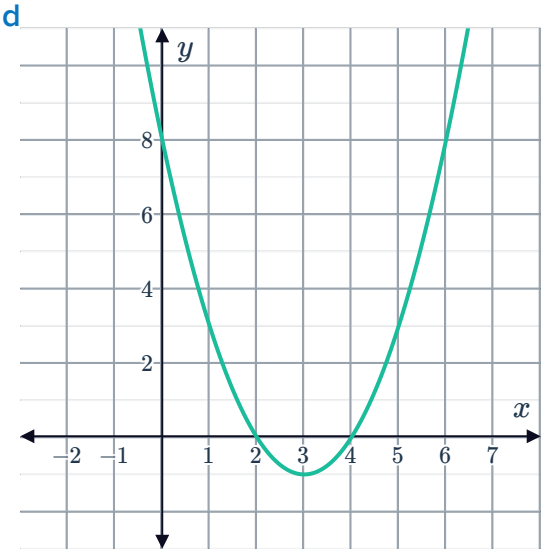
26

a Concave up **b** Less steep **c** $(3, 0)$ **d**



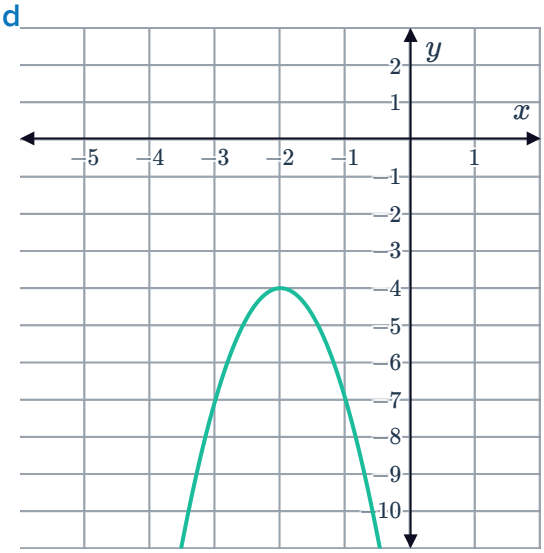
27

a $x = 4, 2$ **b** $y = 8$ **c** $(3, -1)$ **d**



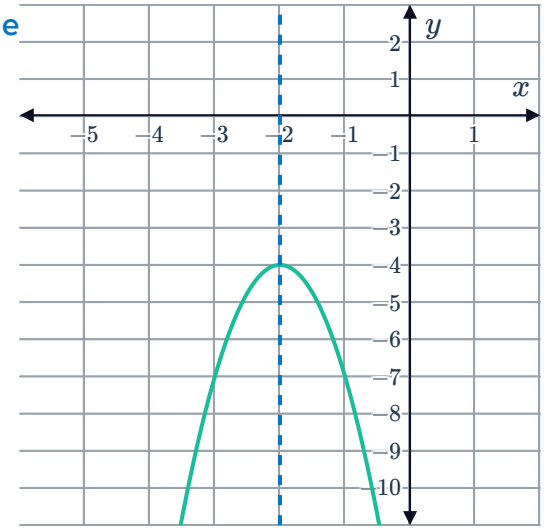
28

a $(-2, -4)$ **b** $x = -2$ **c** $y = -7$ **d**



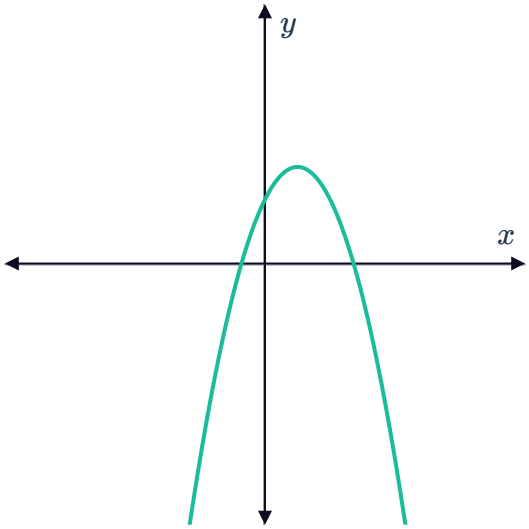
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e

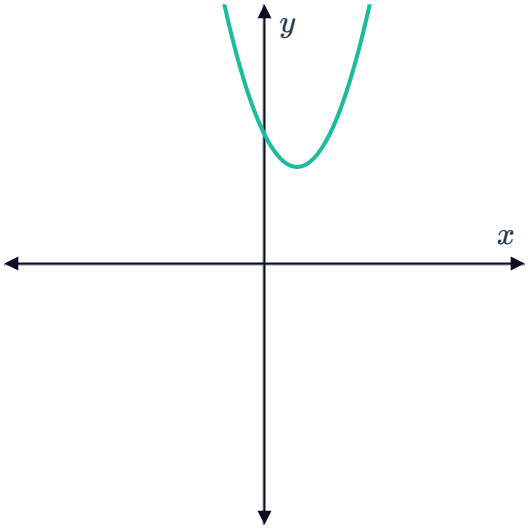


29

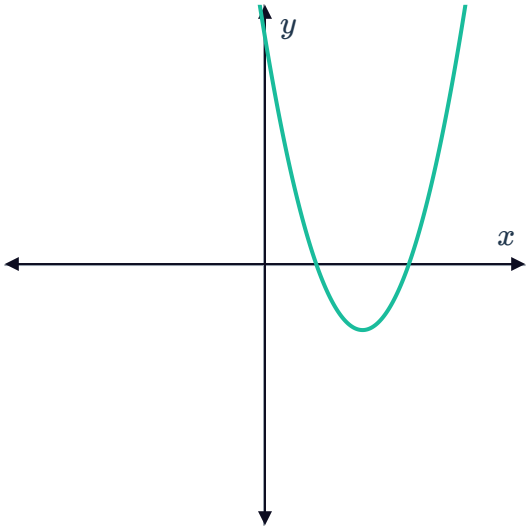
a



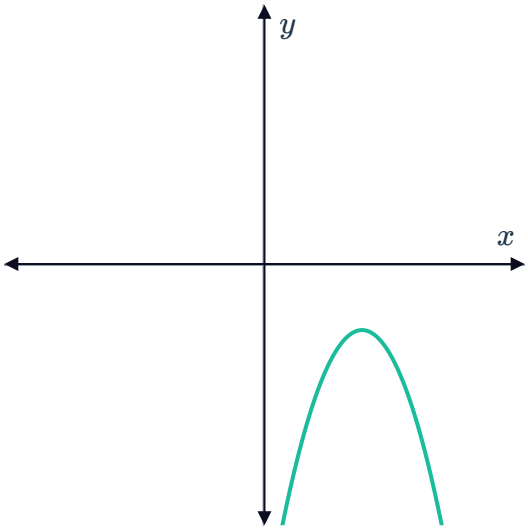
b



c



d



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30

a 8

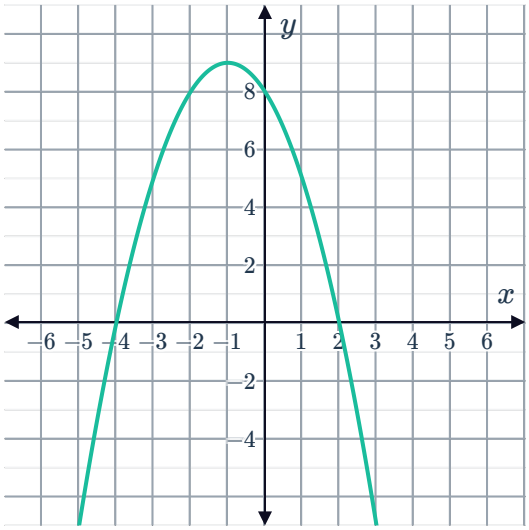
b $x = 2, -4$

c

x	-5	-3	-1	1	3
y	-7	5	9	5	-7

d $(-1, 9)$

e



31

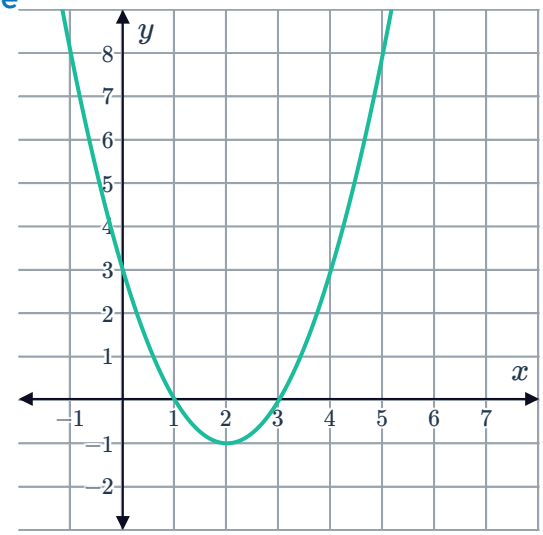
a $y = 3$

b $x = 3, 1$

c $x = 2$

d $(2, -1)$

e



32

a $y = 0$

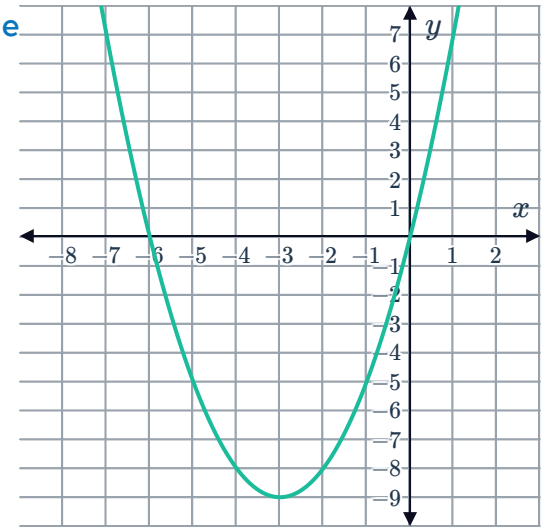
b $x = 0, -6$

c $x = -3$

d $(-3, -9)$

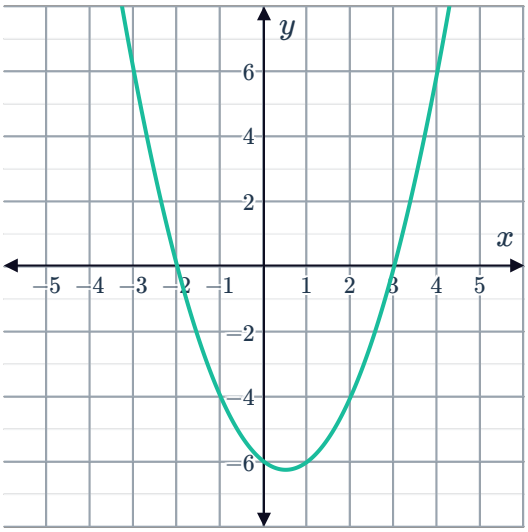
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e

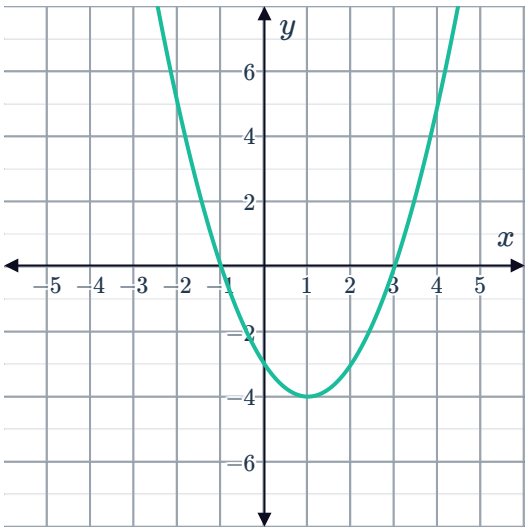


33

a

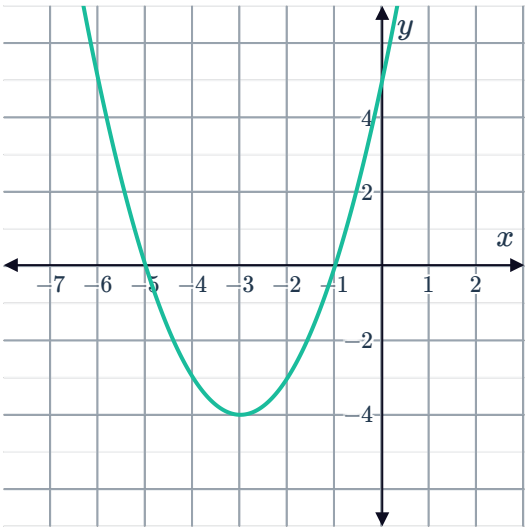


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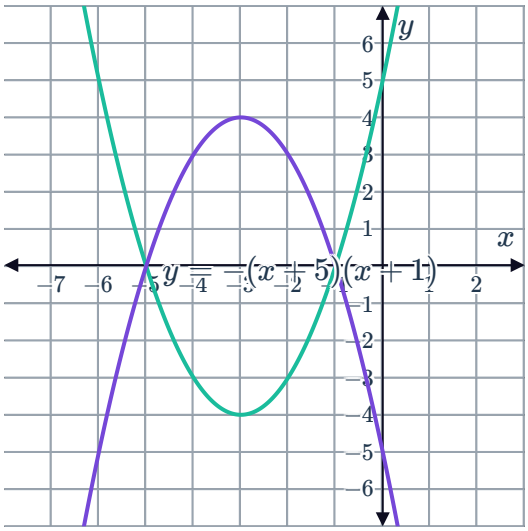


34

a



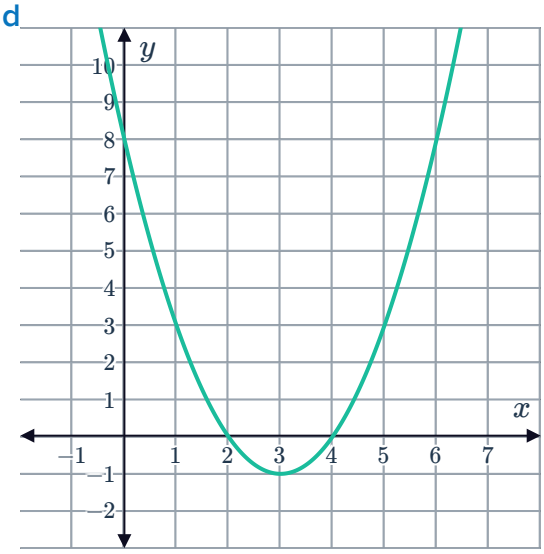
b



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35

a $(x-2)(x-4)$ **b** $x=2,4$ **c** $(3,-1)$ **d**



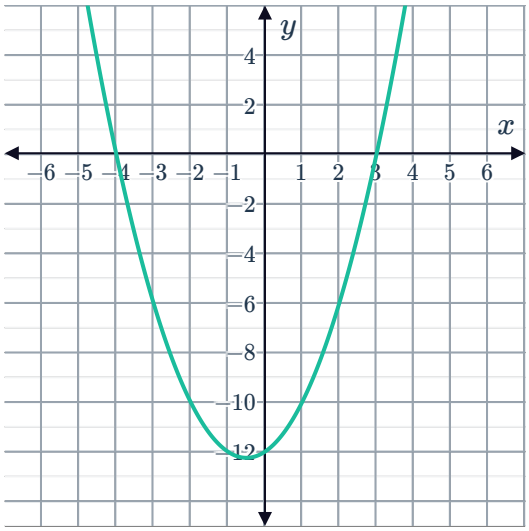
36

a $y=(x+4)(x-3)$ **b** $x=-4,3$

c $y=-12$ **d** $x=-\frac{1}{2}$

e $\left(-\frac{1}{2},-\frac{49}{4}\right)$ **f** Concave up

g



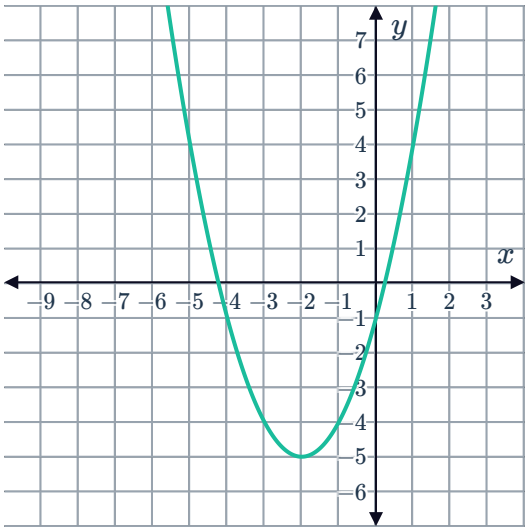
37

a $y=-1$ **b** $(-2,-5)$

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c Concave up

d

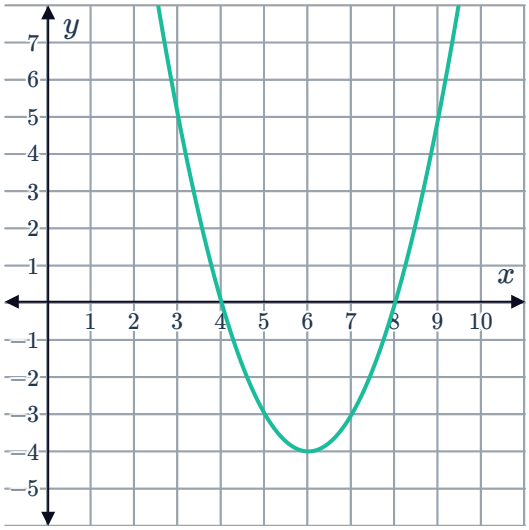


38

a $x = 8, 4$

b $(6, -4)$

c

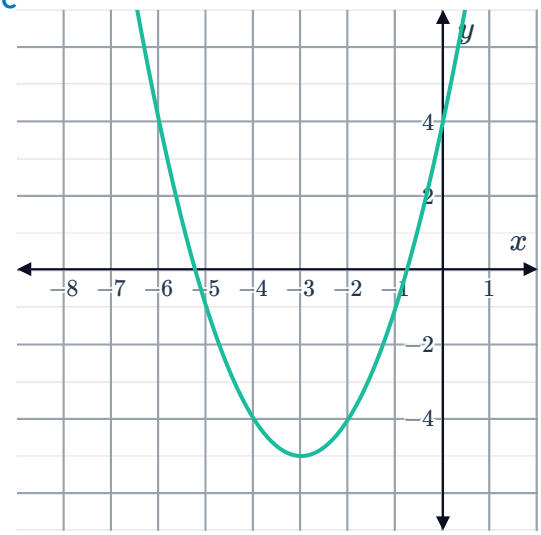


39

a $x = -3$

b -5

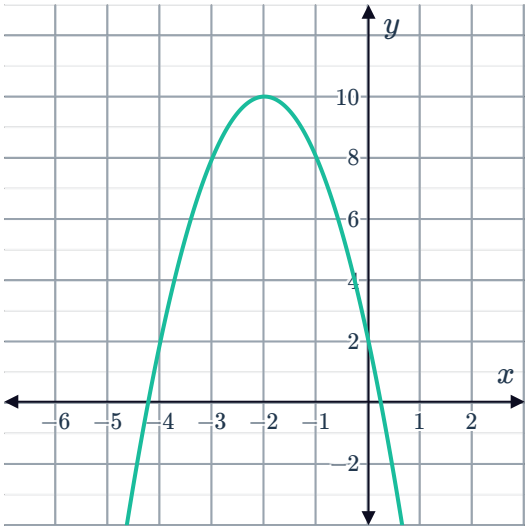
c



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40 **a** $(-2, 10)$

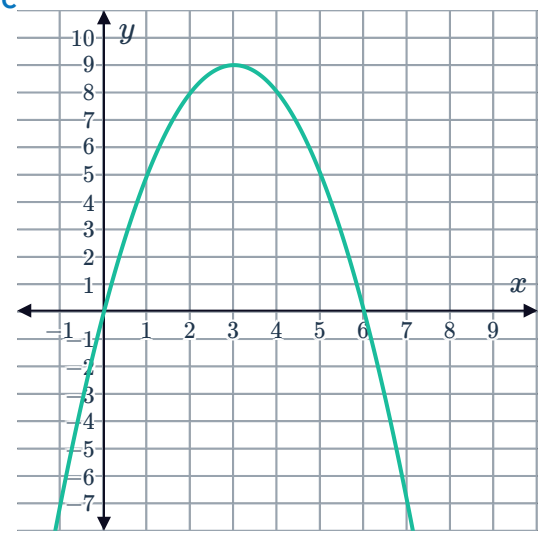
b



41

a $x = 0, 6$ **b** $(3, 9)$

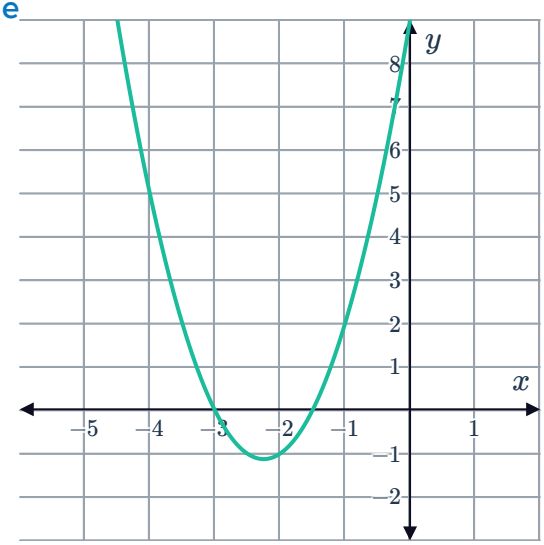
c



42

a $x = -\frac{3}{2}, -3$ **b** $y = 9$ **c** $x = -\frac{9}{4}$ **d** $y = -\frac{9}{8}$

e



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43

a **i** $(8, 7)$ **ii** $(0, 263)$ **b** **i** $(-6, 4)$ **ii** $(0, -140)$

44

a $(0, -12)$ **b** No **c** $x < 0$

45

a $x = -3.1$ **b** -16.61

46

a $x = 2, 6$ **b** $y = -24$ **c** $x = 4$ **d** $y = 8$

47

a $x = 1.55$ **b** $y = 2.95$ **c** Yes **d** $x = 3.58, -0.47$

48

a $x = 2.75$ **b** $y = -9.10$ **c** Yes **d** $x = -0.42, 5.91$

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